

Questions with Answer Keys

#MathBoleTohMathonGo

Q1 - Order, Degree and Formation of DE

If order and degree of the differential equation $\sqrt[3]{\frac{dy}{dx}} - 4\frac{d^2y}{dx^2} - 7x = 0$ are a and b , then the value of $(a + b)$ is

- (1) 3
- (2) 4
- (3) 5
- (4) 6

Q2 - Order, Degree and Formation of DE

The order of differential equation whose general solution is given by $y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{x+c_5}$ (where, c_1, c_2, c_3, c_4 and c_5 are arbitrary constants), is

- (1) 5
- (2) 4
- (3) 3
- (4) 2

Q3 - Order, Degree and Formation of DE

The differential equation of the family of curves represented by $y = a + bx + ce^{-x}$ (where, a, b and c are arbitrary constants) is

- (1) $y''' = y'$
- (2) $y''' + y'' = 0$
- (3) $y''' - y'' + y' = 0$
- (4) None of these

Q4 - Order, Degree and Formation of DE

The differential equation of the family of circles with the fixed radius 5 units and centre on the line $y = 2$, is

- (1) $(x - 2)(y')^2 = 25 - (y - 2)^2$
- (2) $(y - 2)(y')^2 = 25 - (y - 2)^2$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

$$(3) (y-2)^2(y')^2 = 25 - (y-2)^2$$

$$(4) (x-2)^2(y')^2 = 25 - (y-2)^2$$

Q5 - Variable Separable Form and Reducible to Variable Separable Form

The solution of $xy \frac{dy}{dx} = \frac{1+y^2}{1+x^2} (1+x+x^2)$ is

$$(1) \sqrt{1+y^2} = Ce^{\tan^{-1} x}$$

$$(2) 1+y^2 = Cxe^{\tan^{-1} x}$$

$$(3) \sqrt{1+y^2} = Cx$$

$$(4) \sqrt{1+y^2} = Cxe^{\tan^{-1} x}$$

Q6 - Variable Separable Form and Reducible to Variable Separable Form

The solution of $\sin^{-1}\left(\frac{dy}{dx}\right) = (x+y)$ is

$$(1) y = \tan(x+y) - \sec(x+y) + C$$

$$(2) x = \tan(x+y) - \sec(x+y) + C$$

$$(3) y = \log[\tan(x+y)] + C$$

(4) None of the above

Q7 - Variable Separable Form and Reducible to Variable Separable Form

The solution of differential equation $y \log y + xy' = 0$, where $y(1) = e$, is

$$(1) x(\log y) = 1$$

$$(2) xy(\log y) = 1$$

$$(3) (\log y)^2 = 2$$

(4) None of these

Q8 - Variable Separable Form and Reducible to Variable Separable Form

If $y = y(x)$ and $\frac{2+\sin x}{y+1} \cdot \frac{dy}{dx} = -\cos x$, $y(0) = 1$, then $y\left(\frac{\pi}{2}\right)$ equals

$$(1) 1/3$$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(2) $\frac{2}{3}$

(3) $-\frac{1}{3}$

(4) 1

Q9 - Variable Separable Form and Reducible to Variable Separable Form

At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P with respect to additional number of workers is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then new level of production of item is

(1) 2500

(2) 3000

(3) 3500

(4) 4500

Q10 - Variable Separable Form and Reducible to Variable Separable Form

Let $f : \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ (the set of real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right) = 1$. Then, the value of $\int_{1/2}^1 f(x)dx$ lies in the interval

(1) $(2e - 1, 2e)$

(2) $(e - 1, 2e - 1)$

(3) $\left(\frac{e-1}{2}, e - 1\right)$

(4) $\left(0, \frac{e-1}{2}\right)$

Q11 - Homogeneous Form and Reducible to Homogeneous Form

If $x \frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of equation is

(1) $y \log\left(\frac{x}{y}\right) = Cx$

(2) $x \log\left(\frac{y}{x}\right) = Cy$

(3) $\log\left(\frac{y}{x}\right) = Cx$

(4) $\log\left(\frac{x}{y}\right) = Cy$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

Q12 - Homogeneous Form and Reducible to Homogeneous Form

A curve passes through the point $(1, \pi/6)$. Let the slope of the curve at each point (x, y) be

$\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$. Then, the equation of the curve is

(1) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$

(2) $\operatorname{cosec}\left(\frac{y}{x}\right) = \log x + 2$

(3) $\sec \frac{2y}{x} = \log x + 2$

(4) $\cos \frac{2y}{x} = \log x + \frac{1}{2}$

Q13 - Homogeneous Form and Reducible to Homogeneous Form

The solution of $(3xy + y^2) dx + (x^2 + xy) dy = 0$ is

(1) $x^2 (y^2 + 2xy) = C$

(2) $y^2 (y^2 + 2xy) = C$

(3) $y^2 + x^3 y = C$

(4) $y^2 (y^2 - 2xy) = C$

Q14 - Homogeneous Form and Reducible to Homogeneous Form

The solution of $xy \cdot \log\left(\frac{x}{y}\right) dx + \left\{y^2 - x^2 \log\left(\frac{x}{y}\right)\right\} dy = 0$ is

(1) $2x^2 \log\left(\frac{x}{y}\right) - x^2 = 4y^2 \cdot \log\left(\frac{C}{x}\right)$

(2) $2x^2 \log\left(\frac{y}{x}\right) - x^2 = 4y^2 \cdot \log\left(\frac{C}{x}\right)$

(3) $x^2 \log\left(\frac{y}{x}\right) - x^2 = 4y^2 \cdot \log\left(\frac{C}{x}\right)$

(4) None of the above

Q15 - Homogeneous Form and Reducible to Homogeneous Form

The solution of $\frac{dy}{dx} = \frac{2x-y+3}{x+2y+4}$ is

(1) $y^2 - x^2 + xy + 4y - 3x + 1 = C$

(2) $y^2 - x^2 - xy - 4y + 3x - 1 = C$

Questions with Answer Keys

#MathBoleTohMathonGo

(3) $x^2 - y^2 + xy + 4y - 3x - 1 = C$

(4) $y^2 - x^2 + xy + 4y - 3x - 1 = C$

Q16 - Homogeneous Form and Reducible to Homogeneous Form

The solution of $(2x^2 + 3y^2 - 7)x \cdot dx - (3x^2 + 2y^2 - 8)y \cdot dy = 0$ is

(1) $(x^2 + y^2 + 3) = C(x^2 - y^2 - 1)^4$

(2) $(x^2 + y^2 - 3)(x^2 + y^2 + 1) = C$

(3) $(x^2 + y^2 - 3) = (x^2 - y^2 - 1)^5 \cdot C$

(4) None of the above

Q17 - Homogeneous Form and Reducible to Homogeneous Form

The solution of $\frac{dy}{dx} + \frac{\cos x(3 \cos y - 7 \sin x - 3)}{\sin y(3 \sin x - 7 \cos y + 7)} = 0$ is

(1) $(\cos y - \sin x - 1)^2(\sin x + \cos y - 1)^5 = C$

(2) $(\cos y + \sin x - 1)^2(\sin x - \cos y + 1)^5 = C$

(3) $(\cos y + \sin x + 1)^2(\sin x + \cos y + 1)^5 = C$

(4) None of the above

Q18 - Linear Differential Equations and Bernoulli's Equation

If $y(t)$ is a solution of $(1+t)\frac{dy}{dt} - ty = 1$ and $y(0) = -1$, then $y(1)$ is equal to

(1) $\frac{1}{2}$

(2) $e + \frac{1}{2}$

(3) $e - \frac{1}{2}$

(4) $-\frac{1}{2}$

Q19 - Linear Differential Equations and Bernoulli's Equation

Let $p(x)$ be a polynomial such that $p(1) = 0$ and $\frac{d}{dx}[p(x)] > p(x)$ for all $x \geq 1$, then

(1) $p(x) > 0, \forall x > 1$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(2) $p(x) < 0, \forall x > 1$

(3) $p(x) \geq 0, \forall x > 1$

(4) $p(x) \leq 0, \forall x > 1$

Q20 - Linear Differential Equations and Bernoulli's Equation

The solution of differential equation $x^2 \frac{dy}{dx} \cdot \cos\left(\frac{1}{x}\right) - y \cdot \sin\left(\frac{1}{x}\right) = -1$, where $y \rightarrow -1$, as $x \rightarrow \infty$, is

(1) $y = \sin \frac{1}{x} - \cos \frac{1}{x}$

(2) $y = \frac{x+1}{x \cdot \sin(1/x)}$

(3) $y = \cos \frac{1}{x} + \sin \frac{1}{x}$

(4) $y = \frac{x+1}{x \cos(1/x)}$

Q21 - Linear Differential Equations and Bernoulli's Equation

The solution of differential equation $(1 + y^2) + (x - e^{\tan^{-1} y}) \cdot \frac{dy}{dx} = 0$ is

(1) $xe^{2 \tan^{-1} y} = e^{\tan^{-1} y} + C$

(2) $(x - y) = Ce^{-\tan^{-1} y}$

(3) $2xe^{\tan^{-1} y} = e^{2 \tan^{-1} y} + C$

(4) $xe^{\tan^{-1} y} = \tan^{-1} y + C$

Q22 - Linear Differential Equations and Bernoulli's Equation

The solution of differential equation $(2x - 10y^3) \frac{dy}{dx} + y = 0$ is

(1) $x + y = Ce^{2x}$

(2) $y^2 = 2x^3 + C$

(3) $xy^2 = 2y^5 + C$

(4) $x(y^2 + xy) = 0$

Q23 - Linear Differential Equations and Bernoulli's Equation

Consider the differential equation $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If $y(1) = 1$, then x is

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(1) $1 - \frac{1}{y} + \frac{e^{1/y}}{e}$

(2) $4 - \frac{2}{y} - \frac{e^{1/y}}{e}$

(3) $3 - \frac{1}{y} + \frac{e^{1/y}}{e}$

(4) $1 + \frac{1}{y} - \frac{e^{1/y}}{e}$

Q24 - Linear Differential Equations and Bernoulli's Equation

Let $f(x)$ be differentiable on the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$, for each

$x > 0$. Then, $f(x)$ is

(1) $\frac{1}{3x} + \frac{2x^2}{3}$

(2) $\frac{-1}{3x} + \frac{4x^2}{3}$

(3) $-\frac{1}{x} + \frac{1}{x^2}$

(4) $\frac{1}{x}$

Q25 - Linear Differential Equations and Bernoulli's Equation

If $y(x)$ satisfies the differential equation $y' - y \tan x = 2x \sec x$ and $y(0) = 0$, then

(1) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$

(2) $y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$

(3) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{9}$

(4) $y'\left(\frac{\pi}{3}\right) = 4\pi + \frac{2\pi^2}{3\sqrt{3}}$

Q26 - Exact Form

The solution of $xdy + ydx + 2x^3dx = 0$ is

(1) $xy + x^4 = c$

(2) $x^2y + \frac{1}{2}x^4 = e$

(3) $xy + \frac{1}{2}x^4 = C$

(4) $x^2y + x^4 = C$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

Q27 - Exact Form

The solution of $x dx + y dy = a(x^2 + y^2) dy$ is

(1) $\log(x^2 + y^2) = 2ay + C$

(2) $x^2 + y^2 = 2ay + C$

(3) $\log(x^2 - y^2) = 2ay + C$

(4) $x^2 - y^2 = 2ay + C$

Q28 - Exact Form

The solution of $y dx - x dy + (1 + x^2) dx + x^2 \cdot \sin y dy = 0$ is

(1) $\frac{y}{x} + \frac{1}{x} - x + \cos y + C = 0$

(2) $\frac{y}{x} - \frac{1}{x} - x + \cos y + C = 0$

(3) $y + \frac{1}{x} - x + \cos y + C = 0$

(4) $y - \frac{1}{x} - x + \cos y + C = 0$

Q29 - Exact Form

The solution of $(1 + x\sqrt{x^2 + y^2}) dx + (-1 + \sqrt{x^2 + y^2}) y dy = 0$ is

(1) $2x + y^2 + \frac{2}{3}(x^2 + y^2)^{3/2} = C$

(2) $2x - y^2 + \frac{2}{3}(x^2 + y^2)^{3/2} = C$

(3) $x - y^2 + (x^2 + y^2)^{3/2} = c$

(4) $x^2 - y^2 + \frac{2}{3}(x^2 + y^2)^{3/2} = C$

Q30 - Exact Form

The solution of $\frac{xdy}{x^2 + y^2} = \left(\frac{y}{x^2 + y^2} - 1\right) dx$ is

(1) $\tan^{-1}(y/x) = -x + C$

(2) $\tan^{-1}(y/x) = x + C$

(3) $\sin^{-1}(y/x) = -x + C$

(4) $\sin^{-1}(y/x) = x + C$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

Q31 - Exact Form

The solution of $\{y(1+x^{-1}) + \sin y\}dx + \{x + \log x + x \cos y\}dy = 0$ is

(1) $xy + x \sin y + y \log y = C$

(2) $xy + y \sin y + \log x = C$

(3) $xy + x \sin y + y \log x = c$

(4) $x + xy \sin y + y \log x = C$

Q32 - Exact Form

The solution of $x \cos(y/x) \cdot (ydx + xdy) = y \sin(y/x)(xdy - ydx)$ is

(1) $xy = C \sin(y/x)$

(2) $xy = C \cos(y/x)$

(3) $xy = C \sec(y/x)$

(4) $xy = C \tan(y/x)$

Q33 - Orthogonal Trajectory

The orthogonal trajectory of $y = cx^2$, where c is an arbitrary constant, is

(1) $x^2 - 2y^2 = C_1$

(2) $x^2 + y^2 + x = C_1$

(3) $x^2 + 2y^2 = 2C_1$

(4) $x^2 - 2y^2 = 2C_1$

Q34 - Orthogonal Trajectory

The orthogonal trajectory of $x^2 + y^2 - 2cx = 0$ (where, c is a parameter) is

(1) $x^2 + y^2 + y = c$

(2) $x^2 + y^2 + x = 0$

(3) $x^2 + 2y^2 + Cy = 0$

(4) $x^2 + y^2 + Cy = 0$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

Answer Key

Q1 (3)	Q2 (3)	Q3 (2)	Q4 (3)
Q5 (4)	Q6 (2)	Q7 (1)	Q8 (1)
Q9 (3)	Q10 (4)	Q11 (3)	Q12 (1)
Q13 (1)	Q14 (1)	Q15 (4)	Q16 (3)
Q17 (1)	Q18 (4)	Q19 (1)	Q20 (1)
Q21 (3)	Q22 (3)	Q23 (4)	Q24 (1)
Q25 (1)	Q26 (3)	Q27 (1)	Q28 (1)
Q29 (2)	Q30 (1)	Q31 (3)	Q32 (3)
Q33 (3)	Q34 (4)		